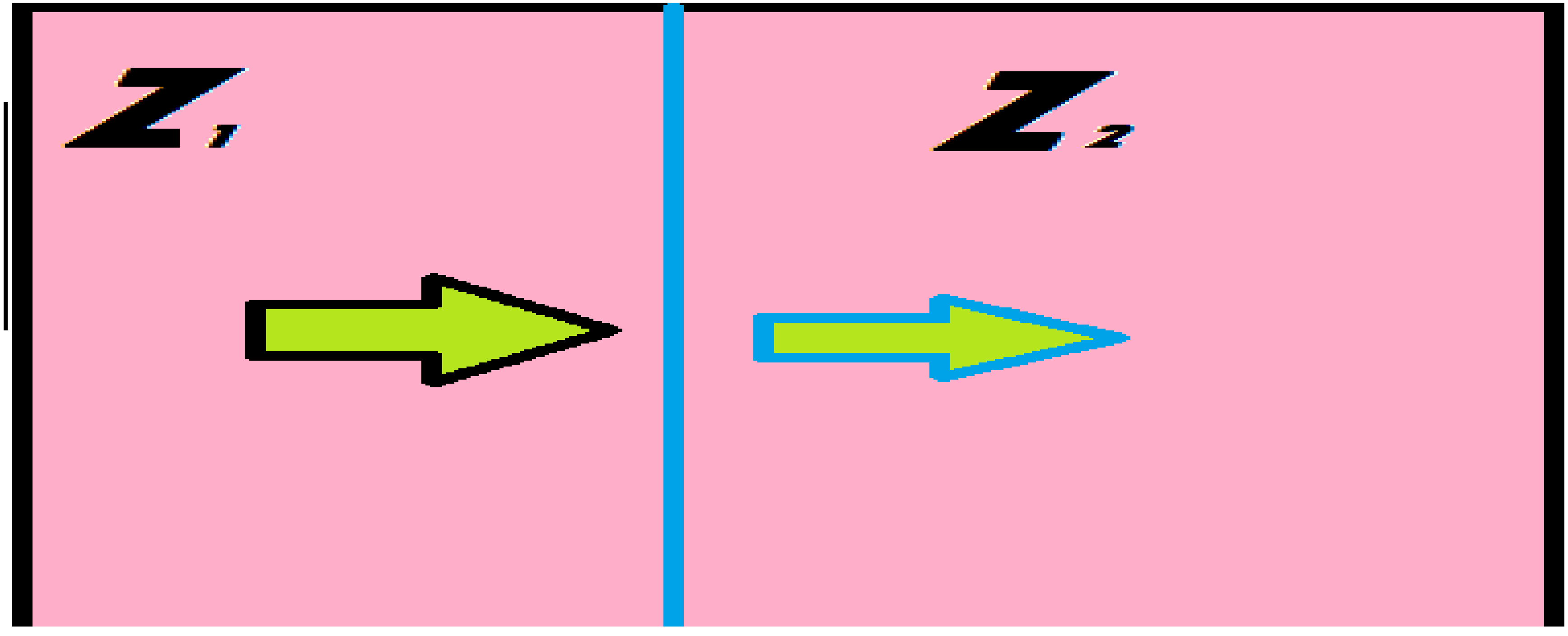


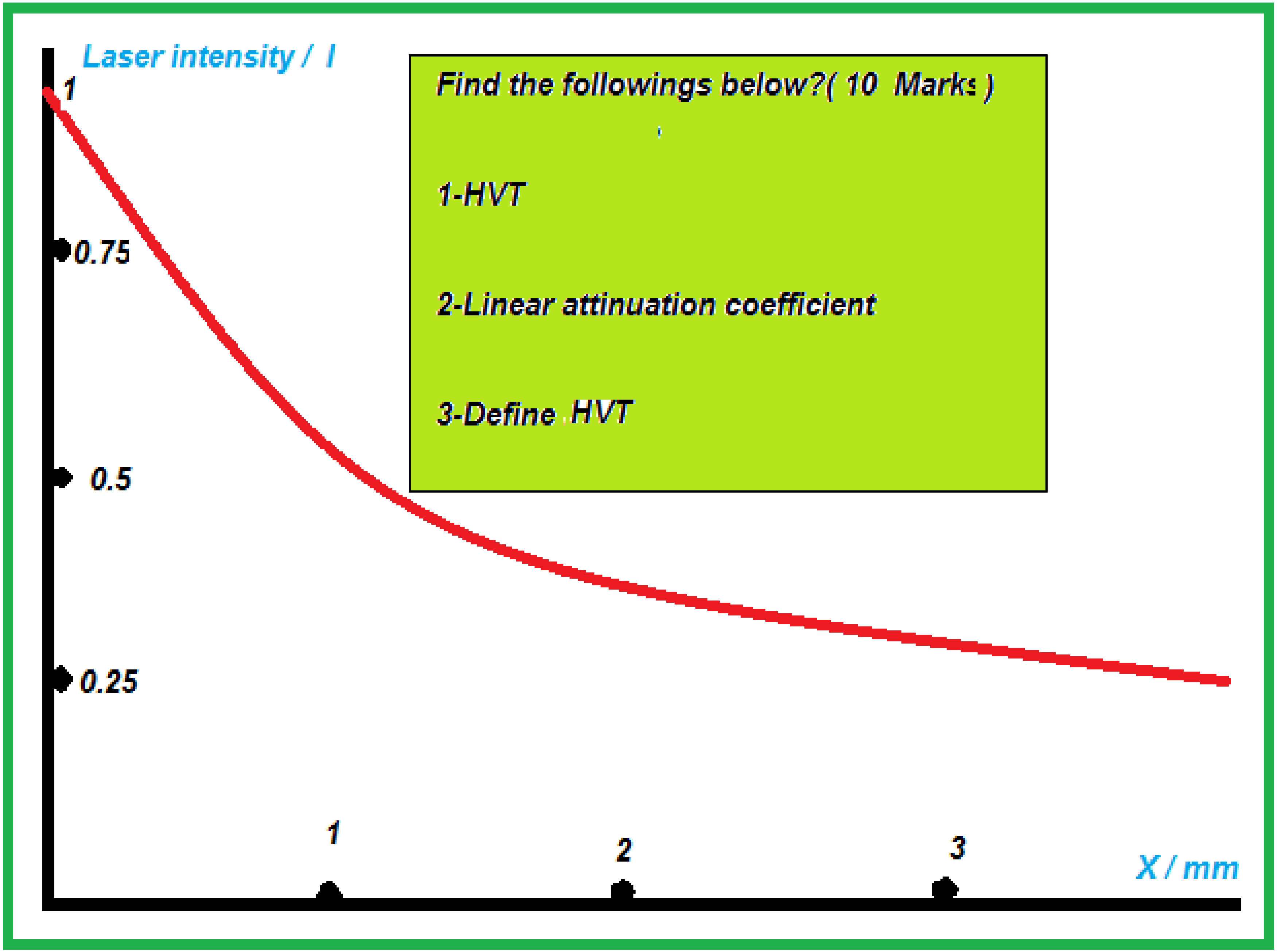
1-If z1=z2 , write the formula of Reflection or Transmission ?

2-Define Z ?

3-Write the units of Z?

Determination the linear attenuation coefficient and half value thickness of LASER

- Half Value Thickness; $HVT=0.693/\mu$
- Define HVT ?
- Find linear attenuation coefficient?



Find the followings below?(10 Marks)

1-HVT

2-Linear attinuation coefficient

3-Define HVT

Lasers in medicine

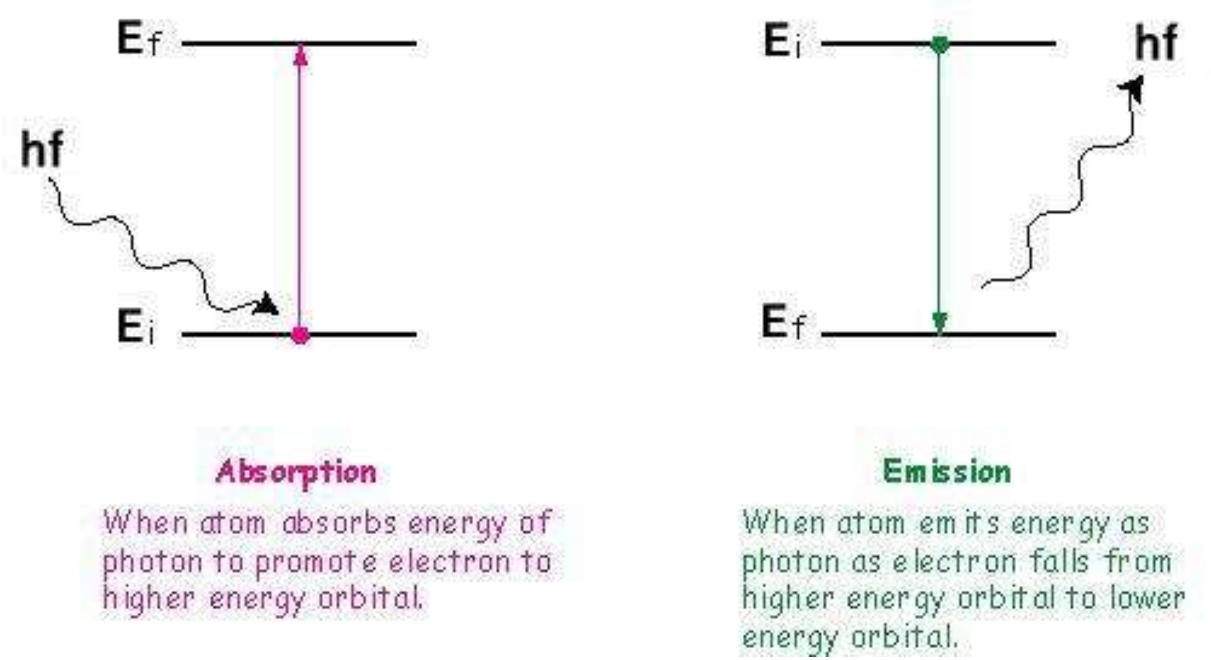
INTRODUCTION;

While the basic theory for lasers was proposed by Albert Einstein in 1917, the first successful laser was not made until 1960scientists T.H Maiman produced a laser beam from a ruby crystal . Since 1960 scientists have made many types of lasers using gases and liquids as well as solids.

Review of concepts

- Energy levels of electrons bound to atoms is discrete
  - These finite, discrete values are unique to the atom (helps identify atom using Spectroscopy)
- An atom at a higher energy state emits light (photons) to return to a lower energy state
  - Atoms are unstable at higher energy states
  - Wavelength of light emitted is proportional to the finite energy difference
- Finite number of energy levels implies finite wavelengths

Absorption / Emission of Photons



- Photon energy

$$E_{photon} = E_{final} - E_{initial}$$

- The frequency f of the emitted photon must is determined by

$$E_{photon} = hf$$

LASER

- Light Amplification by Stimulated Emission of Radiation
- Laser beams properties are;
  - Coherent
  - Intense
  - single frequency
  - Narrow
- Can be generated in several ways
- Wide applications

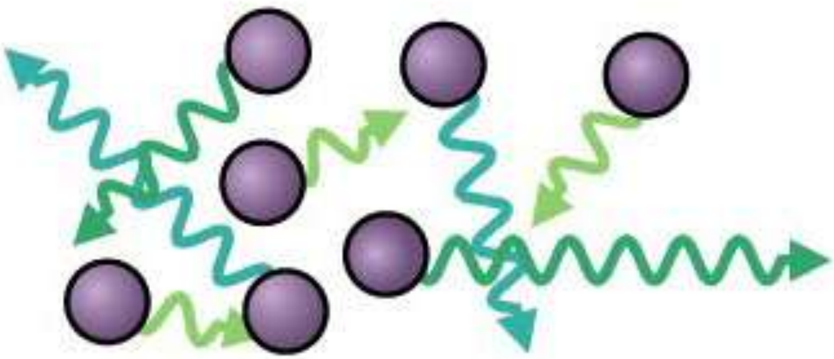
Types of lasers

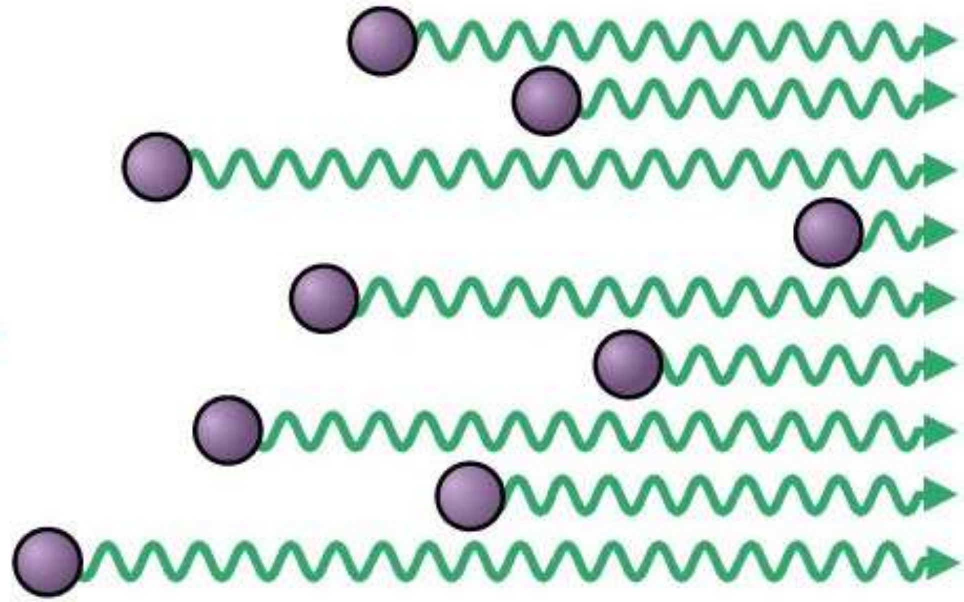
- Gas lasers
- Chemical lasers
- Dye lasers
- Metal-vapor lasers
- Solid-state lasers
- Semiconductor lasers
- Free electron lasers
- Raman lasers
- Nuclear pumped lasers

Laser generation process

- Three steps
  - Energy absorption (pumping)
    - Absorbs light energy / electrical energy
    - Atoms in material move to higher energy state
  - Spontaneous emission
    - Material emits light to move from higher (unstable) energy state to lower (more stable) energy state spontaneously
  - Stimulated emission
    - Spontaneously emitted light triggers other high energy atoms to lose energy and emit light
    - **This step produces a highly coherent laser beam**

Coherent and Incoherent radiation

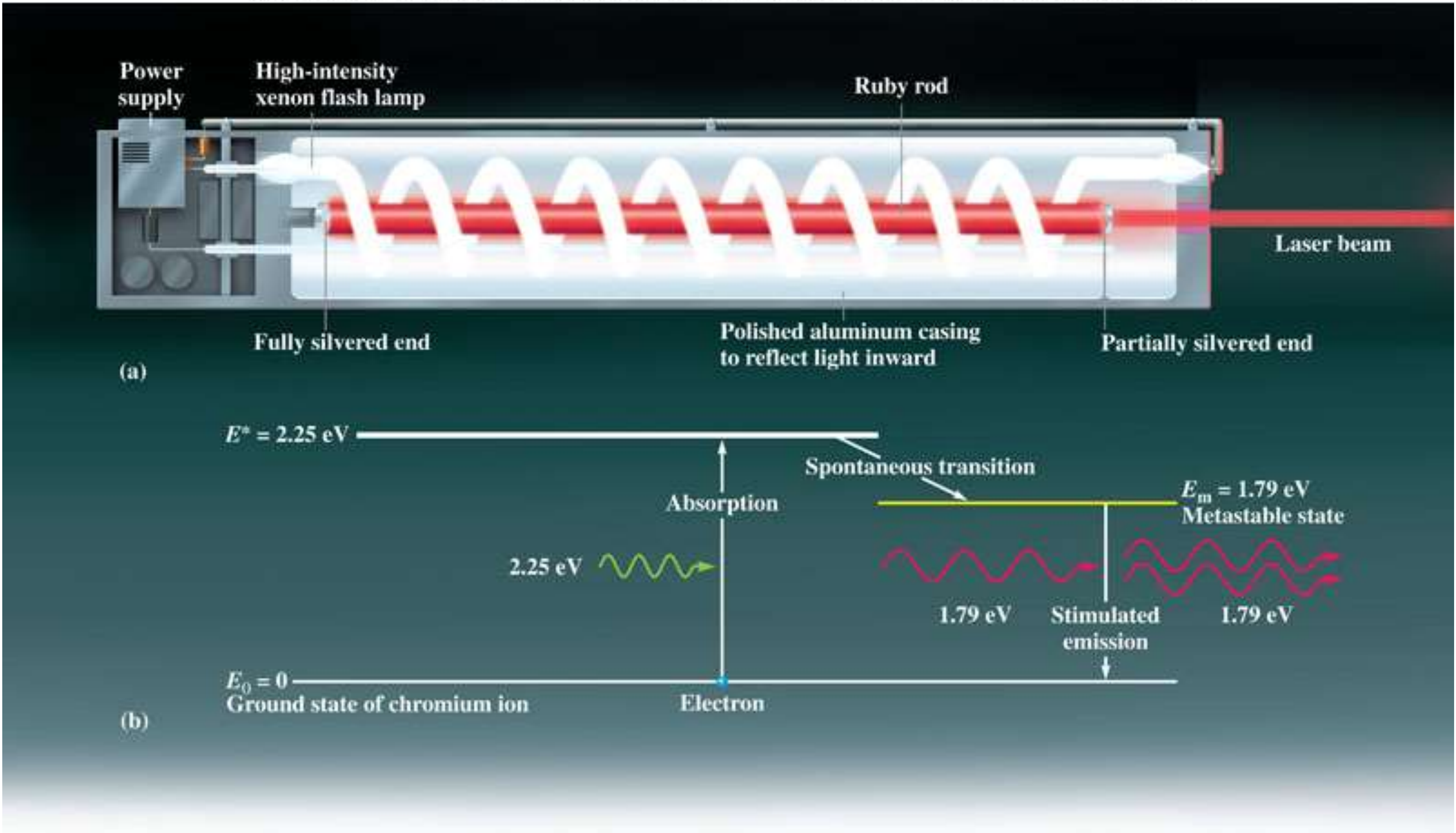




Ruby laser

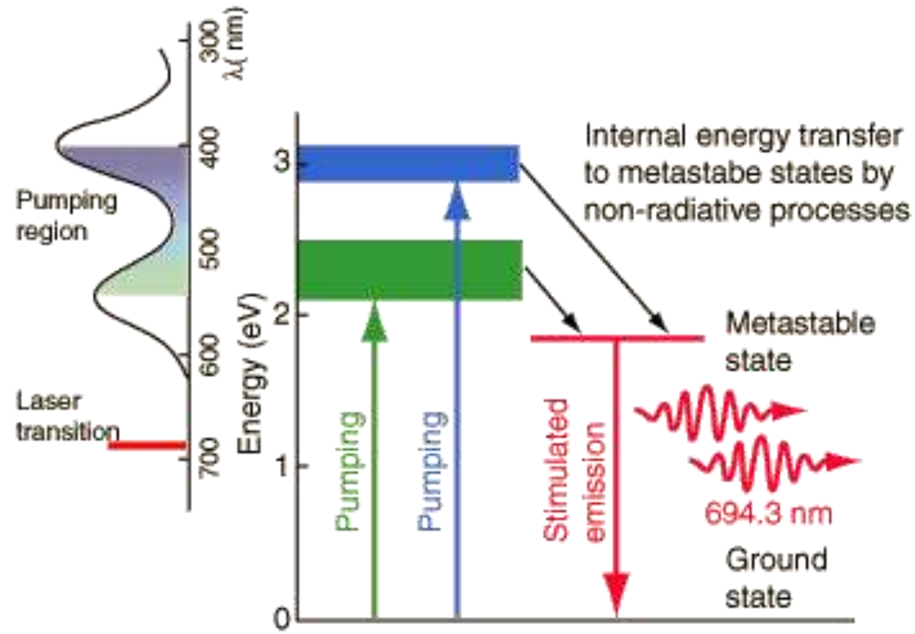
- First laser ever made (1960 by Ted Maiman, Hughes Aircraft company)
- Emits pulses of visible light at a wavelength of 694.3 nm
- Used in holography, non-destructive testing, hair removal
- Three level laser
- Active laser gain medium is a synthetic ruby rod
  - Mostly made up of Al_2O_3 and doped with Cr^{3+} ions

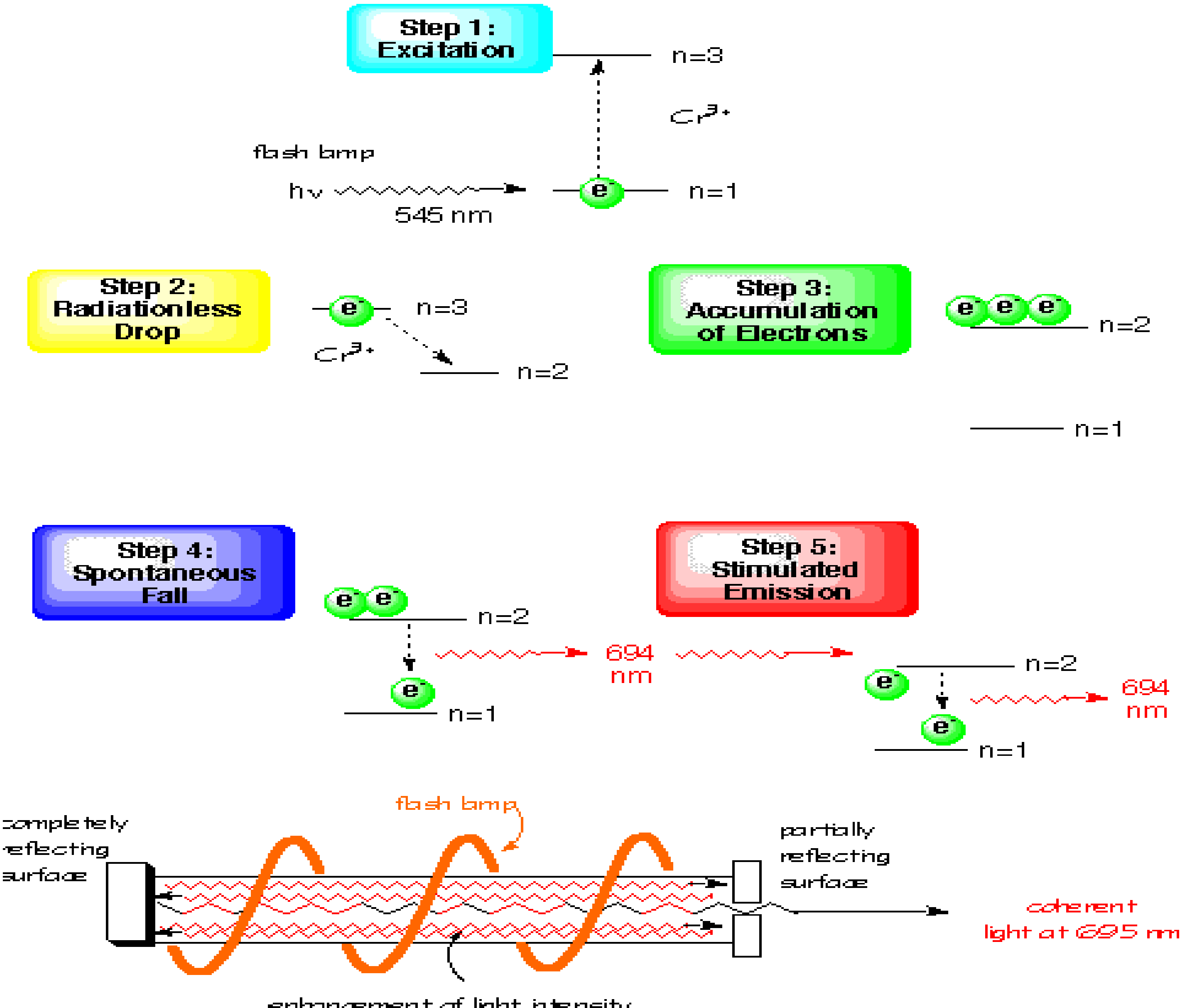
Ruby Laser


Power supply
High-intensity xenon flash lamp
Ruby rod
Laser beam
Fully silvered end
Polished aluminum casing to reflect light inward
Partially silvered end
(a)
E* = 2.25 eV
Spontaneous transition
Absorption
Em = 1.79 eV
Metastable state
2.25 eV
1.79 eV
Stimulated emission
1.79 eV
E0 = 0
Ground state of chromium ion
Electron
(b)

Ruby Laser working

- Step 1: Excitation
- Step 2: Radiation Loss
- Step 3: Electron Accumulation (population inversion)
- Step 4: Spontaneous fall
- Step 5: Stimulated emission (radiation generation).

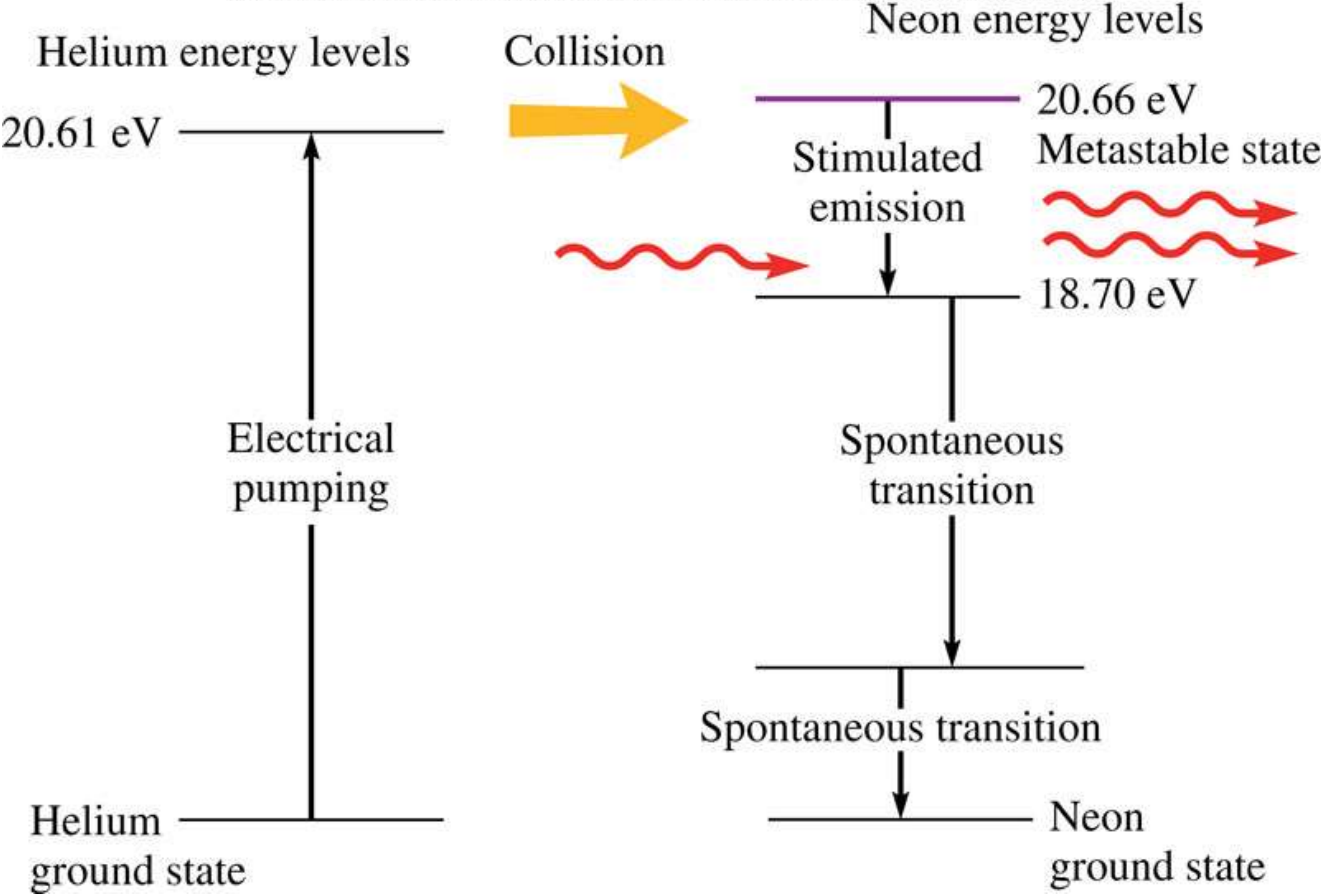




Step 1: Excitation
n=3
Cr^{3+}
flash lamp
hν
545 nm
e⁻
n=1
Step 2: Radiationless Drop
e⁻
n=3
Cr^{3+}
n=2
Step 3: Accumulation of Electrons
e⁻ e⁻ e⁻
n=2
n=1
Step 4: Spontaneous Fall
e⁻ e⁻
n=2
694 nm
e⁻
n=1
Step 5: Stimulated Emission
n=2
e⁻
694 nm
e⁻
n=1
flash lamp
completely reflecting surface
partially reflecting surface
coherent light at 695 nm

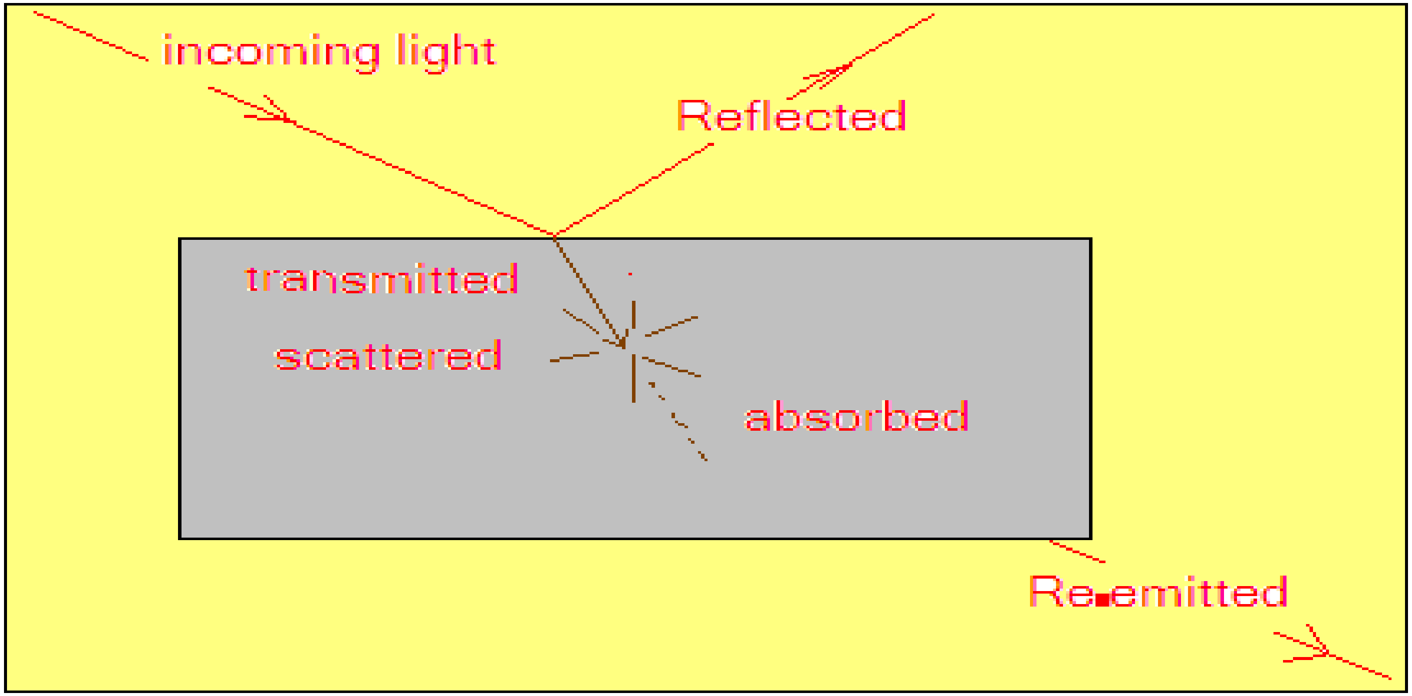
Helium-Neon laser

- Small gas laser emitting 632.8 nm light
- Gain medium is a mixture of helium and neon combined in a low pressure cavity
- Pumped by 1000 V, 5-100 mA signal applied between cathode and anode
- Electrical signal pumps helium from ground to excited states
- High energy helium atoms collide with neon atoms in ground state till population inversion occurs
- This triggers lasing action


Helium energy levels
Collision
Neon energy levels
20.61 eV
20.66 eV
Metastable state
Stimulated emission
18.70 eV
Electrical pumping
Spontaneous transition
Spontaneous transition
Helium ground state
Neon ground state

How laser interacts with body tissues

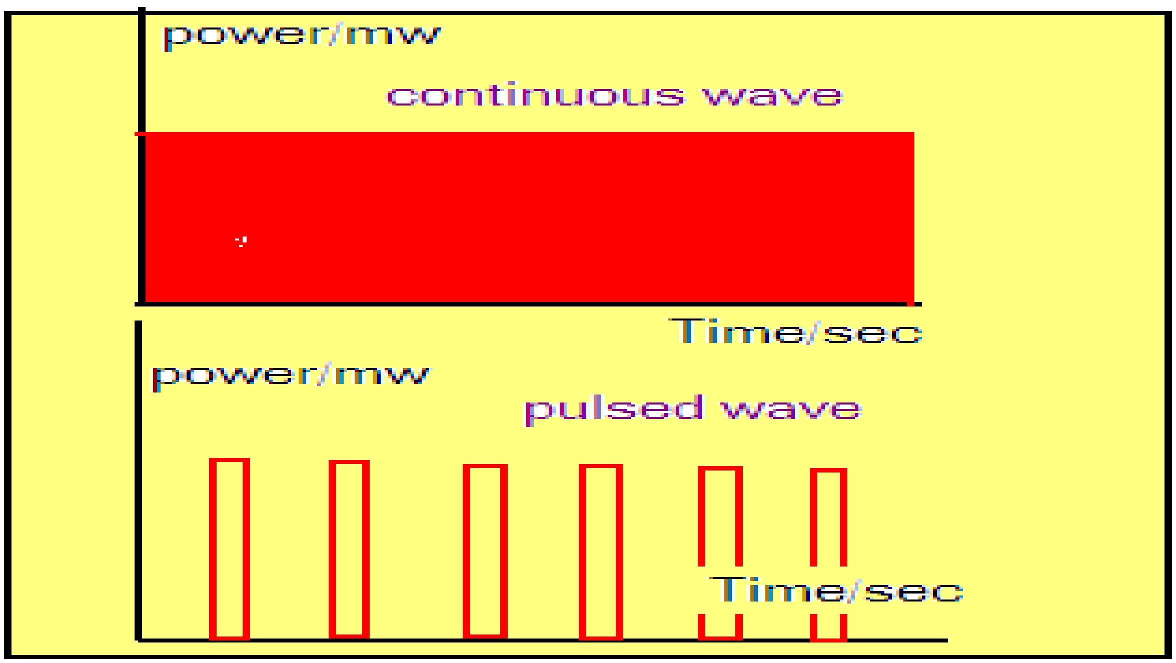
- We are principally interested in how light of various wavelengths interacts with the body tissues, this is because laser surgery and light therapy both involve the transfer energy from light to human tissue though several mechanisms Figure below shows that;


incoming light
Reflected
transmitted
scattered
absorbed
Re-emitted

Lasers can operate in one of two modes, shown in Figure below;

1-Continuous wave (CW).

2-Pulsed LASER.



Applications of lasers

- Medical
  - Eye surgery (LASIK,…)
  - Dental procedures
  - Cosmetic surgery
  - Imaging
  - Laparoscopy

LASER in dermatology.

-LASER in surgery as knife.

-LASER in wound and fractures healing.

-LASER used in killing of tumor cells.

- LASER used in amputation in diabetic disease.

Application of LASER IN DENTISTRY

- Surgery of soft tissues of mouth and jaw.
- Used as a drill.
- Teeth bleaching.
- Reshape teeth and pain less.

There are some equations applied in problems in medicine.

- C=wavelength x frequency.
- E(J)=P(watt)X T(sec)
- I =P(watt)/A(cm^2)
- F(Fluence(J/cm^2)=I(watt/cm^2) X T (sec)

LASIK

- Laser-Assisted in Situ Keratomileusis
- Used to correct myopia, hypermetropia and astigmatism
- Uses an ultraviolet excimer laser (193 nm)
- Shape of the cornea is changed using the laser

FIBER OPTIC APPLICATIONS IN MEDICINE (ENDOSCOPES AND LAPAROSCOPES)

Endoscopes; defined as the device that illumination and viewing internal cavity of body by special flexible tubes, for diagnosis of diseases and imaging the needed internal part of the body.

Medical endoscopes go by specific names that are determined by which parts of the body each scope is used to image. A partial listing is given in Table below, many different types exist, each designed for specific applications, Endoscopes are used for examining the inside of the body , Type of scope   Used for imaging of

Endoscope   (Hollow organs ;the gastrointestinal tract and eyes).

Laparoscope   (Abdominal cavity, gynecological surgery, gall bladder removal).

Bronchoscope (Bronchial passage of the lung) .

Cystoscope   (Urinary bladder)

Colonoscopy (Large intestine).

Sigmoidoscope (Outer region of the large). intestine.

Thorascope chest cavity.

Laryngoscope Larynx.

Angioscope Blood vessels.

Arthroscope interior of joints; knee, ankle.
shoulder, elbow.

Simple problem on lasers

Conceptual Example 28.5

Photocoagulation

An argon ion laser is used to repair vascular abnormalities and fissures in the retina of the eye in a process known as photocoagulation. Laser light *absorbed* by the tissue raises its temperature until proteins become coagulated, forming the scar tissue that repairs the split. The principal wavelengths emitted by the argon laser are 514 nm and 488 nm. (a) What are the photon energies for these wavelengths? (b) What are the colors associated with these two wavelengths? (See Fig. 22.7.) Are both wavelengths effective on blood vessels?

Strategy The energy of a photon is related to its wavelength by $E = hc/\lambda$. Section 22.4 lists the colors of the visible spectrum and the associated wavelengths. A wavelength is effective if it is strongly absorbed.

Solution and Discussion (a) The photon energies are

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV}\cdot\text{nm}}{514 \text{ nm}} = 2.41 \text{ eV}$$

and

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV}\cdot\text{nm}}{488 \text{ nm}} = 2.54 \text{ eV}$$

(b) The color associated with 514 nm is green and with 488 nm is blue. Both wavelengths are effective on blood vessels because red blood vessels reflect red and absorb radiation of other colors.

Conceptual Practice Problem 28.5 Ruby Laser and Blood

Would red light from a ruby laser be effective on blood and thus useful in the treatment of vascular abnormalities?